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The Climate Crisis

A report published in 2021 surveyed 10,000 individuals aged 16-25 across multiple countries, revealing that 59% expressed significant worry about climate change, with 84% reporting at least moderate concern. Furthermore, more than 45% of respondents acknowledged that their feelings about climate change negatively impacted their daily lives and functioning (Hickman, Caroline, et al.). These findings highlight a concerning trend among younger generations, indicating a substantial proportion of kids and young adults grapple with the tangible effects of climate anxiety. While it's important to acknowledge the reality of climate change, it's worth examining the assumption that the perceived crisis, particularly concerning carbon dioxide emissions, warrants such pervasive negative impacts on daily life, as observed in the experiences of a significant portion of the surveyed population.

In his 1968 book '*The Population Bomb*,' biologist Paul Ehrlich famously predicted global famines and social upheaval due to unsustainable population growth, a prediction that failed to materialize as critics noted the absence of his predicted catastrophes. Contrary to Ehrlich's forecast, the period following his predictions saw global economies thriving, with products becoming increasingly affordable across various sectors, all while the world's population continued to grow. In response to

Ehrlich's predictions, economist Julian Simon proposed a wager to Ehrlich, challenging him to select five commodities and predict that their prices would increase over time. Ehrlich chose copper, chromium, tin, tungsten, and nickel, with the wager set to conclude ten years after 1980. Ultimately, when the wager concluded in 1990, Ehrlich lost as the prices of all selected commodities had declined, serving as a notable intellectual counterpoint to Ehrlich's earlier warnings. Ehrlich was just one among many who have introduced alarmist predictions into the discourse surrounding Earth's climate, and humanity's role in shaping it, that has no doubt contributed in some amount to the climate anxiety observed in the experiences of a significant portion of the surveyed population.

Alarming statements about climate change continued to emerge from influential figures. Former Vice President Al Gore famously warned in 2009 of a 75% chance that the entire north polar ice cap could be ice-free within five to seven years, a claim misinterpreted from its source. In 2008, Al Gore similarly predicted the disappearance of the North polar ice cap within five years. Climate activist Greta Thunberg stirred controversy by tweeting in 2018, and then deleting years later, that a top climate scientist warned of humanity's extinction within five years if fossil fuel use wasn't halted. In 2021, U.S. climate envoy under the Biden administration, John Kerry asserted that scientists had given humanity twelve years to avert the worst climate consequences, leaving nine years remaining. Such extreme alarmism from intellectuals, politicians, and activists undoubtedly contributes to climate anxiety among the younger generation. This raises questions about the blurring of facts and alarmism in the discourse surrounding climate change.

Carbon dioxide currently has an atmospheric concentration of approximately 425 parts per million (ppm) as measured at the Mauna Loa Observatory in Hawaii. While data from this observatory dates back only to its establishment in 1958, earlier information can be obtained through alternative methods (US Department of Commerce).

One research team, led by M. Rubino et al., was able to retrieve data from ice cores in Antarctica. As snow compacts and forms ice layers over time, it traps air bubbles representing the Earth's atmosphere and corresponding time periods with remarkable accuracy. Their findings indicate that carbon dioxide concentrations remained relatively stable at around 280 ppm from 1000 A.D. to 1800 A.D. However, since the industrialization period, human activity has led to a consistent growth in CO₂ levels, with concentrations rising to today's value. While this nearly one and a half times increase may seem substantial, its impact on global temperatures has not mirrored this trend to the degree one might think it would.

One group's research, led by Schmidt et al., highlighted the significant role of carbon dioxide in the Earth's greenhouse effect, accounting for approximately 20 percent of the total effect, with water vapor contributing 50 percent and clouds 25 percent. Their research calculated conditions wherein carbon dioxide was doubled in concentration and where it was removed from the atmosphere altogether, revealing the contrasting effects on temperature and the greenhouse effect. Remarkably, the complete removal of carbon dioxide would result in a temperature drop of approximately -35 degrees Celsius, leading to an 'ice-covered planet.' Conversely, doubling carbon dioxide concentrations would yield a comparatively modest increase in temperature, estimated at 2.7 degrees Celsius. This discrepancy shows carbon dioxide has a nonlinear effect on temperature and

therefore as CO₂ concentrations rise, even larger increases in CO₂ levels are needed to produce equivalent temperature changes. Though a drastic rise in CO₂ concentrations is occurring, the associated temperature rise is not as quantitatively substantial.

As previously established, carbon dioxide accounts for 20 percent of the greenhouse effect, contributing to global warming (Schmidt et al.). Regulatory bodies like the Environmental Protection Agency (EPA) have categorized carbon dioxide as a pollutant greenhouse gas due to its role in climate change (Environmental Protection Agency). However, carbon dioxide also possesses beneficial qualities. While high concentrations can be toxic to humans and are a byproduct of industrial activities, CO₂ is essential for plant growth, a fact often overlooked in discussions about its environmental impact. Categorizing CO₂ solely as a pollutant may overlook its crucial role in supporting plant life, upon which human existence depends. Therefore, it may be a potential mistake to overlook the vital role of CO₂ in sustaining ecosystems and biodiversity.

Plants undergo photosynthesis to produce energy which requires water and carbon dioxide as an input and yields sugars and the oxygen we breathe as an output (Piao, Shilong, et al.). It stands to reason that if there is more carbon dioxide to make food for the plants then there will be more plant growth. This was the case for a study by Taylor and Wolfram that used data from NASA's Orbiting Carbon Observatory-2 satellite which revealed that even a small increase of 1 ppm in CO₂ corresponds to substantial yield boosts: 0.4% for corn, 0.6% for soybeans, and 1% for wheat, respectively. Furthermore, the researchers used these findings to analyze historical crop yield data dating back to 1940, demonstrating that carbon dioxide had served as the primary driver of yield

growth. Despite these benefits, the designation of carbon dioxide as a pollutant by the EPA risks overshadowing its crucial role in supporting crop production and global food security for an ever growing population. The fertilization effect of carbon dioxide is not limited to just that of crops, but also extends to that of other plants as well.

During the early study of climate, as carbon dioxide concentrations and emissions data had been collected long enough to study, scientists saw a mysterious trend that couldn't be accounted for as it was understood at the time. In one paper that looked at the sources and sinks of carbon dioxide, researchers had found that the known emissions and sinks of carbon dioxide do not match. In other words, the humans were producing more carbon dioxide than what was being measured in the atmosphere and oceans. (Tans, P. P., et al.). They proposed that the discrepancy must be as a result of a terrestrial sink that was removing significant amounts of carbon dioxide from the atmosphere. As the data compiled over time and scientists were able to study this inconsistency, they were able to pinpoint the cause. A study led by Zhu et al. that utilized satellite data to analyze plant growth found that from 1982 to 2009 there was an increase in vegetation over 25 to 50 percent of the global vegetated area. They further analyzed the factors contributing to this surge in plant growth and determined that carbon dioxide was accountable for roughly 70 percent of this increase. Most recently, research conducted by Chen et al. using NASA's satellite data, has provided compelling evidence of a remarkable phenomenon: from 2000 to 2017, the total leaf area on Earth increased by 5.4 million square kilometers—a size equivalent to that of the Amazon rainforest. This unprecedented greening of the planet underscores the vital role of terrestrial ecosystems in sequestering carbon dioxide and promoting environmental health.

Elevated carbon dioxide levels offer more than just direct biomass generation. They also influence plants' water use efficiency, particularly in arid and semi-arid environments. Plants absorb carbon dioxide through pores in their leaves called stomata, which also regulate the release of water vapor through transpiration. When carbon dioxide levels are higher, plants can reduce the aperture of their stomata, thereby minimizing water loss. This enhanced water use efficiency enables plants to thrive in water-limited environments, allowing them to colonize and flourish in semi-arid regions. (Piao, Shilong, et al.). The importance of elevated CO₂ levels extends beyond direct benefits to plants as it also plays a crucial role in the greening of Earth and it aids crop growth in arid conditions, thereby contributing to ecosystem resilience and global food security.

The relationship between carbon dioxide concentration and plant productivity is crucially important, as demonstrated by the adverse effects of decreasing CO₂ levels on plant growth and function. Throughout Earth's history, the planet has experienced multiple glacial periods characterized by significantly lower temperatures. During these periods, the solubility of carbon dioxide in water increases, leading to a decline in atmospheric CO₂ levels and posing challenges for plant growth (Gerhart and Ward). Studies, such as those conducted by Gerhart and Ward, have revealed that during the late glacial maximum around 18,000 to 20,000 years ago, carbon dioxide levels plummeted to 180 to 190 parts per million ppm. This is among the lowest concentrations recorded in the evolutionary history of plants, and as levels drop to as low as 150 ppm, some plants lose their ability to flower and reproduce. (Cunniff, J., et al.). This data underscores the critical role of carbon dioxide in sustaining plant productivity and the functioning of

ecosystems. Furthermore, the impact of low carbon dioxide concentrations on plant productivity is hypothesized to have been a limiting factor on human advancement. The timing of the end of the last ice age aligns with the emergence of agriculture, a pivotal stage of development in human history. Studies, such as those by Cuniff et al., have demonstrated that reduced carbon dioxide levels had a significant effect on crop growth and water retention, suggesting that agriculture emerged when it did because low CO₂ concentrations posed limitations. This hypothesis is supported by archaeological evidence, as hindcasting methods applied by Kavanagh et al. to past human populations have revealed that improving environmental conditions, including rising CO₂ levels, favored higher local population densities during periods of agricultural emergence in various regions.

The discourse surrounding climate change is complex and often driven by alarmist narratives propagated by influential figures. Exaggerated claims and sensationalist rhetoric contribute to a pervasive sense of fear and anxiety among the general population, leading to negative impacts on mental health and daily functioning. However, it is crucial to critically examine these alarmist narratives and recognize the disconnect between perception and reality. While carbon dioxide is often categorized as a pollutant, scientific evidence reveals its vital role in supporting plant life and promoting agricultural productivity. Alarmist narratives overlook the significant benefits associated with increased CO₂ levels, such as the remarkable greening of the Earth by areas larger than the Amazon rainforest and the enhancement of ecosystem resilience. By adopting a more balanced and evidence-based approach to discussions about climate change, we can

mitigate the negative impacts of climate anxiety and empower individuals to take positive action towards a sustainable future.

The multifaceted role of carbon dioxide in shaping Earth's ecosystems and human development underscores the need for a nuanced understanding of its effects. While regulatory bodies categorize CO₂ as a pollutant greenhouse gas, it is essential to recognize its beneficial qualities, particularly in agriculture and plant growth. Alarmist narratives often overlook the significant greening of the Earth and the remarkable resilience of ecosystems in response to increased CO₂ levels. Moreover, the modest temperature increase associated with doubled CO₂ concentrations highlights the need for a reassessment of crisis-level framing in discussions about climate change. Rather than viewing CO₂ emissions as a looming catastrophe, we should recognize the opportunities for innovation, adaptation, and collaboration in addressing environmental challenges.

The challenges posed by climate change are significant, but they are also accompanied by opportunities for positive change. By critically examining alarmist narratives and recognizing the benefits of increased CO₂ levels, we can adopt a more balanced and evidence-based approach to climate discourse. It is imperative to communicate responsibly about climate change to avoid exacerbating mental health impacts and empower individuals to take meaningful action towards a sustainable future. Let us embrace a perspective that acknowledges both the risks and benefits of rising CO₂ levels, in order to avoid worsening mental health impacts and empower individuals to take positive collective action in addressing environmental problems.

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